

Joseph Robert Livesey

CURRICULUM VITAE

Contact	Address: 4120 11th Ave NE, Apt 306 Seattle, WA 98105 Phone: +1 (715) 505-3347	Email: jlivesey@uw.edu Website: https://jrlivesey.github.io GitHub: https://github.com/jrlivesey
Personal	Pronouns: he/him Citizenship: U.S.A. ORCID: 0000-0003-3888-3753	
Research Interests	Planetary system dynamics, planetary atmospheres, planetary formation, astrobiology, open source software for astronomy.	
Education	B.S. ASTRONOMY & PHYSICS <i>CUM LAUDE</i> University of Washington With Honors in Astronomy, Honors in Physics, Minors in Mathematics & Linguistics	2018–2022
Research Experience	POST-BACCALAUREATE RESEARCHER, University of Washington <i>Supervisor: Rory Barnes</i> Evaluating the long-term evolution and orbital stability of planetary systems using secular and N -body orbital dynamics simulations.	2022–2023
	UNDERGRADUATE RESEARCHER, University of Washington <i>VPLANET Group; Supervisor: Rory Barnes</i> Modeling the orbits and interiors of potentially habitable exoplanets on gigayear timescales. <i>Condensed Matter Experiment; Supervisor: Amal al-Wahish</i> Evaluating the diffusive properties of alkaline earth phosphates through spectroscopy.	2019–2022
Honors and Awards	Phi Beta Kappa Mary Gates Research Scholarship (\$5,000 award) Quarterly Dean's List, UW College of Arts & Sciences ($\times 11$)	Jun 2021 Dec 2020 2019–2022
Teaching Experience	READER/GRADER ASTR 150: The Planets (Instructor: Rory Barnes) ASTR 427: Computational Methods (Instructor: Matt McQuinn) ASTR 101: Astronomy (Instructor: Chris Laws) ASTBIO 115: Life in the Universe (Instructor: Rory Barnes) ASTR 101: Astronomy (Instructors: Oliver Fraser & Suzanne Hawley)	Winter 2023 Autumn 2022 Autumn 2022 Spring 2022 Winter 2022
Service	Scientific Organizing Committee Member, VPLANET Workshop 2	Sep 2022
Professional Development	VPLANET Developers Workshop (<i>virtual workshop</i>)	Jun 2021
Organizations	Phi Beta Kappa UW League of Astronomers UW Research Computing Club	2021–Present 2021–2022 2021–2022
Skills	Programming: Python, Interactive Python, Mathematica (<i>proficient</i>) C, SQL (<i>familiar</i>) Software: Unix, Git, L ^A T _E X (<i>proficient</i>) HTML, CSS (<i>familiar</i>)	

Non-Academic Employment	Library Technician, Odegaard Undergraduate Library	Jul-Sep 2022
	Access Services Student Assistant, Odegaard Undergraduate Library	2018-2022
	Front End Assistant, Costco Wholesale	Jul-Sep 2018
Publications	FIRST-AUTHOR	
	1. "Dynamics of the Proxima Centauri System I: Orbital Stability." Livesey, J. R. & Barnes, R. <i>In prep.</i>	
	CO-AUTHOR	
	2. "History and Habitability of the LP 890-9 Planetary System." Barnes, R., do Amaral, L. N. R., Birky, J., Carone, L., Driscoll, P., Livesey, J. R. , Garcia, R., Gialluca, M. et al. <i>In prep.</i>	
Conference Proceedings	FIRST-AUTHOR	
	1. "Orbital History of the Proxima Centauri System." Livesey, J. R. ,* Barnes, R. & Deitrick, R. (2023 January; accepted). Oral presentation at AAS Meeting 241, Seattle, WA.	
	2. " Orbital and Rotational Evolution of Proxima Centauri b ." Livesey, J. R. * & Barnes, R. (2021 January). Oral presentation at AAS Meeting 237, virtual.	
	CO-AUTHOR	
	3. "History and Habitability of the LP 890-9 Planetary System." Barnes, R.,* do Amaral, L. N. R., Birky, J. et al. (<i>incl. Livesey, J. R.</i>) (2023 January; accepted). Oral presentation at AAS Meeting 241, Seattle, WA.	
	4. " Interdisciplinary Modeling of Planetary Habitability ." Barnes, R.,* Barth, P., Wilhelm, C. et al. (<i>incl. Livesey, J. R.</i>) (2021 March). Oral presentation at Habitable Worlds 2021, virtual.	
	* = presenting author	
Contributed Talks	Dynamics of the Proxima Centauri Planets	2021
	<i>Undergraduate Research Symposium, UW</i>	<i>Virtual</i>
	Orbital and Rotational Evolution of Proxima Centauri b	2020
	<i>AstroFest, UW</i>	<i>Virtual</i>
	Structure and Dynamics Investigation of Newberyite (MgHPO ₄ ·3H ₂ O)	2020
	<i>Undergraduate Research Symposium, UW</i>	<i>Virtual</i>
References	Prof. Rory Barnes, UW Astronomy & Astrobiology	rkb9@uw.edu
	Prof. Eric Agol, UW Astronomy & Astrobiology	agol@uw.edu
	Dr. Amal al-Wahish, UW Physics	wahish@uw.edu
Relevant Coursework	ASTRONOMY	PHYSICS
	Solar System/Exoplanets (321, 419)	Classical Mechanics (121, 329)
	Stellar Astrophysics (322, 421)	Electricity & Magnetism (122, 321, 322)
	Extragalactic Astronomy (323)	Waves & Optics (123)
	High-Energy Astrophysics (423)	Thermodynamics (224)
	Cosmology (425)	Quantum Mechanics (225, 324, 325)
	Programming for Astronomy (300, 302)	Mathematical Methods (227, 228)
	Astrostatistics & Machine Learning (324)	Particle Physics (226)
	Computational Methods (427)	Statistical Mechanics (328)
	Astronomical Data Analysis (480)	Atomic Physics (421)
		Electric Circuits Lab (334)
		Computational Physics Lab (434)
		Physics Honors Seminar (485, 486, 487)
	MATHEMATICS	
	Elementary Calculus (124, 125, 126)	

Differential Equations (307, 309)

Linear Algebra (308)

Vector Calculus (324)

Probability (394; *not taken for grade*)

OTHER

Computational Linguistics (LING 471)

Last updated: December 13, 2022